SLE-3: ARCHITECTURAL DESIGN USING FULL C4 MODEL

Course: 02AML204 – Introduction to Artificial Intelligence

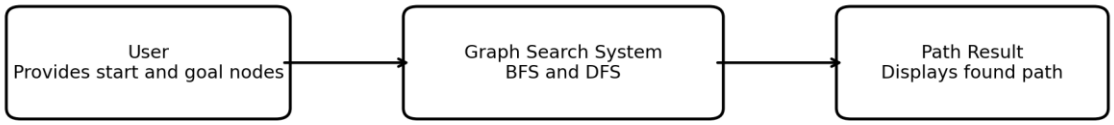
1. System Title and Short Description

Project Title: Graph Search System using BFS and DFS

The Graph Search System finds a path from a start node to a goal node in a graph. It uses Breadth First Search (BFS) and Depth First Search (DFS). BFS explores nodes using a queue, while DFS explores nodes using a stack. The system tracks visited nodes to avoid unnecessary repeated work and displays the path found by each algorithm.

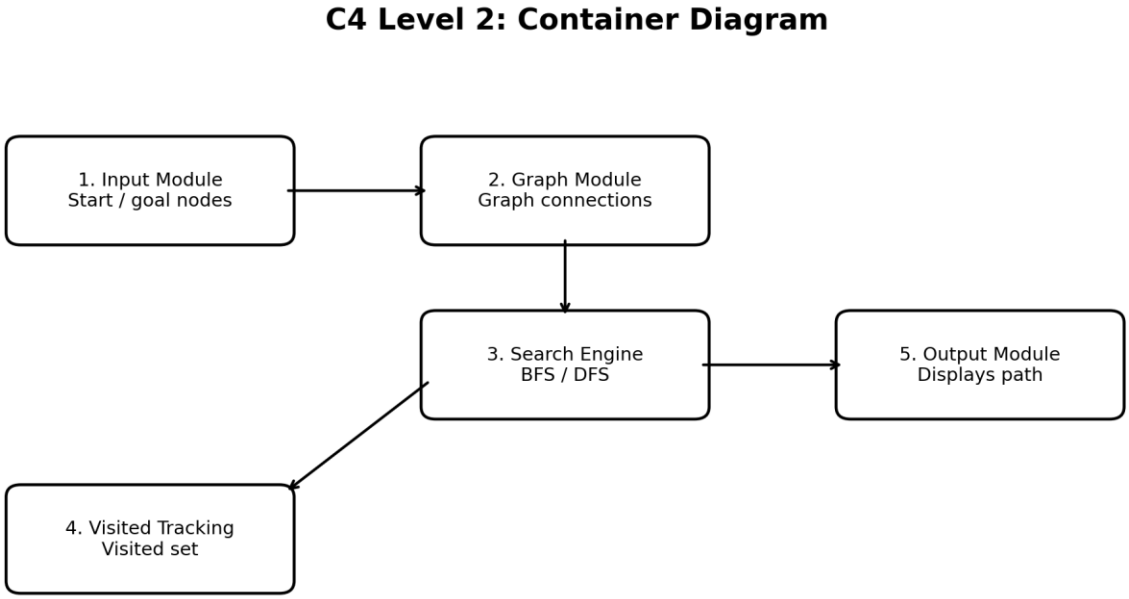
2. Context Diagram (C4 Level 1)

C4 Level 1: Context Diagram



Explanation: The user provides the start and goal nodes. The Graph Search System searches the graph using BFS or DFS. The system then displays the path found. The user interacts with the system through input and output.

3. Container Diagram (C4 Level 2)

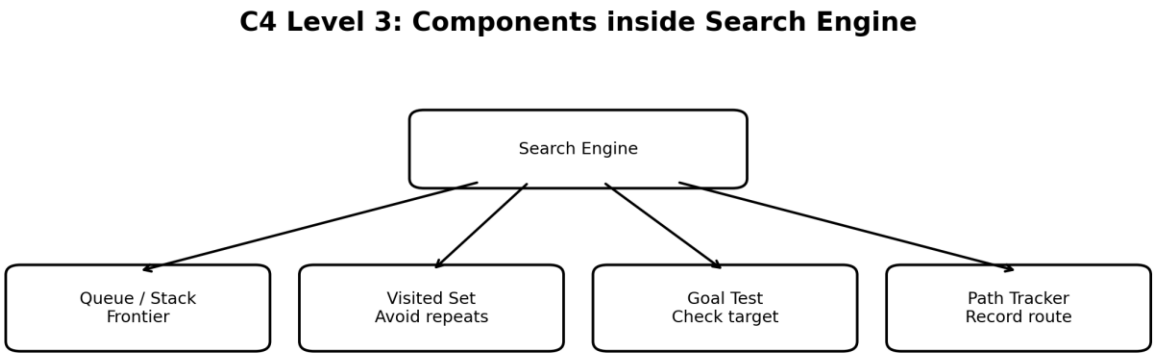


Explanation of containers:

- **Input Module:** Accepts the start node and goal node for the search.
- **Graph Module:** Stores nodes and the connections between them.
- **Search Engine:** Runs the BFS or DFS algorithm to search for the goal.
- **Visited Tracking:** Stores already visited nodes to avoid repeated exploration.
- **Output Module:** Displays the path found by the selected algorithm.

4. Component Diagram (C4 Level 3)

The Search Engine is selected as the main container for this component-level view.



Explanation: The Queue/Stack stores the next nodes to explore. The Visited Set tracks nodes already processed. The Goal Test checks whether the current node is the target. The Path Tracker stores the route from the start node to the goal.

5. Code Level Overview (C4 Level 4)

- **graph —** Stores graph connections as an adjacency list.
- **bfs(start, goal) —** Searches the graph using a queue and returns a path.
- **dfs(start, goal) —** Searches the graph using a stack and returns a path.
- **visited —** Keeps track of nodes already explored.
- **path —** Stores the current route from the start node.
- **print() —** Displays the BFS and DFS paths.

6. Design Decisions

The system is divided into small logical modules so that each part has a clear responsibility. BFS uses a queue, while DFS uses a stack, making it easy to compare their search behaviour. A visited set helps prevent repeated exploration of the same nodes.

7. AI Contribution Note

AI tool used: ChatGPT.

AI helped me understand the four C4 levels, organize the system architecture, and improve the explanations. I reviewed the design, studied the BFS and DFS

algorithms, and ran or checked the Python code and its output. I will make sure I can explain each diagram and component.

8. Conclusion

This assignment helped me understand how software architecture can be shown at four levels, from the overall system to functions and variables. I learned how input, graph data, search logic, visited tracking, and output work together. The C4 Model makes the system easier to explain and maintain.

GitHub / Portfolio Link

Add your repository link here:

https://github.com/pritishinde832-art/SLE3_26uam304/blob/main/AI%20Contribution%20Log%20%E2%80%93%20SLE-3